Experiment-11: Travelling Salesman Problem (TSP) Using Exhaustive Search

Aim : To find the minimum-cost tour that visits every city exactly once and returns to the starting city using the Exhaustive Search method.

Algorithm

1. Read the number of cities and the distance matrix.
2. Fix the first city as the starting city.
3. Generate all possible orders of visiting the remaining cities.
4. Calculate the total distance for each tour, including the return to the starting city.
5. Find the tour with the minimum total distance.
6. Display the optimal tour and its minimum cost.

Python Program

```
from itertools import permutations
n = int(input("Enter number of cities: "))
print("Enter the distance matrix:")
dist = []
for i in range(n):
    row = list(map(int, input().split()))
    dist.append(row)
if n < 2:
    print("At least 2 cities are required.")
else:
    min_cost = float("inf")
    best_path = None
    for path in permutations(range(1, n)):
        tour = (0,) + path + (0,)
        cost = 0
        for i in range(len(tour) - 1):
            cost += dist[tour[i]][tour[i + 1]]
        if cost < min_cost:
```

```
            min_cost = cost
            best_path = tour
    print("Optimal Tour:", " -> ".join(
        str(city + 1) for city in best_path
    ))
    print("Minimum Cost:", min_cost)
```

Input

Enter number of cities: 4

Enter the distance matrix:

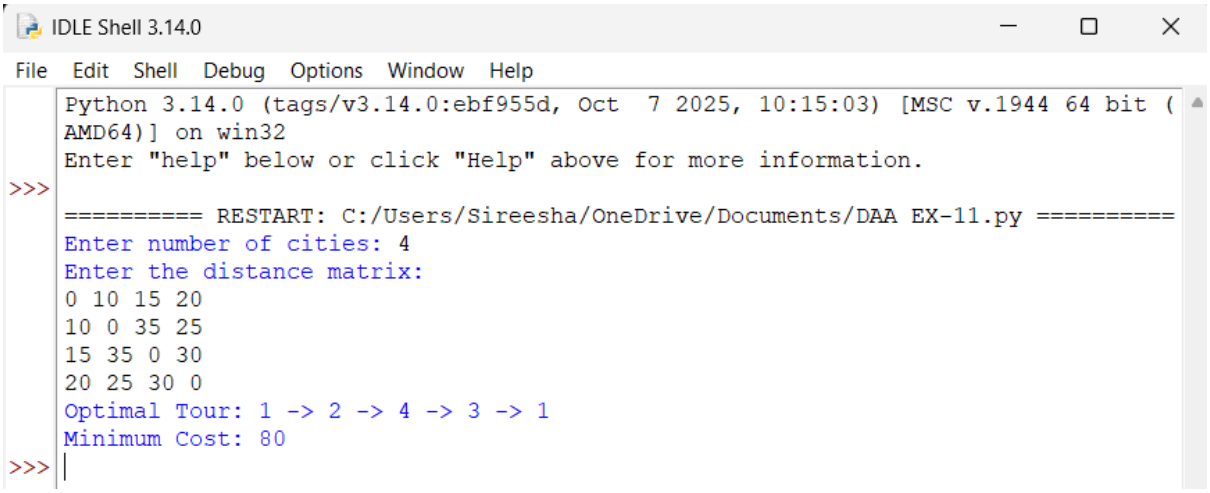
0 10 15 20

10 0 35 25

15 35 0 30

20 25 30 0

Output



Result

Thus, the Travelling Salesman Problem was successfully implemented using the Exhaustive Search method, and the optimal tour with the minimum travel cost was obtained.